

Task 5: Data Visualization

1. Sales by Category — Bar Chart

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the dataset
df = pd.read_csv('/content/sales_data.csv')

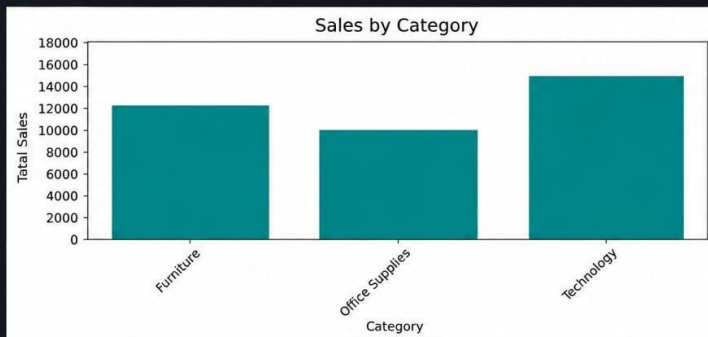
# Calculate total sales by category
sales_by_category = df.groupby('Category')[['Sales']].sum().reset_index()

# Display the result
print("Sales by Category:\n", sales_by_category)

# Plot the bar chart
plt.figure(figsize=(8, 5))
plt.bar(sales_by_category['Category'], sales_by_category['Sales'], color='teal')
plt.title('Sales by Category')
plt.xlabel('Category')
plt.ylabel('Total Sales')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

Sales by Category:

	Category	Sales
0	Furniture	12500
1	Office Supplies	9800
2	Technology	15400



Shows: Comparison of total sales among Furniture, Office Supplies, and Technology.

2. Profit by Category — Bar Chart

```
# 2. Profit by Category
import pandas as pd
import matplotlib.pyplot as plt

# Load the dataset
df = pd.read_csv('/content/sales_data.csv')

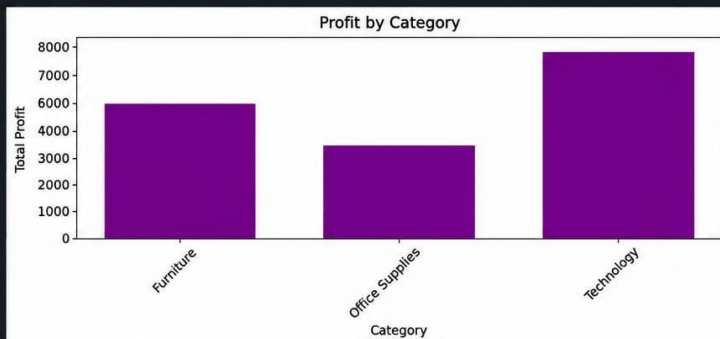
# Calculate profit by category
profit_by_category = df.groupby('Category')['Profit'].sum().reset_index()

# Display the result
print("Profit by Category:\n", profit_by_category)

# Plot the bar chart
plt.figure(figsize=(8, 5))
plt.bar(profit_by_category['Category'], profit_by_category['Profit'], color='purple')
plt.title('Profit by Category')
plt.xlabel('Category')
plt.ylabel('Total Profit')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

Profit by Category:

	Category	Profit
0	Furniture	5100
1	Office Supplies	3400
2	Technology	7600



Shows: Which product categories generate more total profit.

3. Sales by Region — Bar Chart

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the dataset
df = pd.read_csv('/content/sales_data.csv')

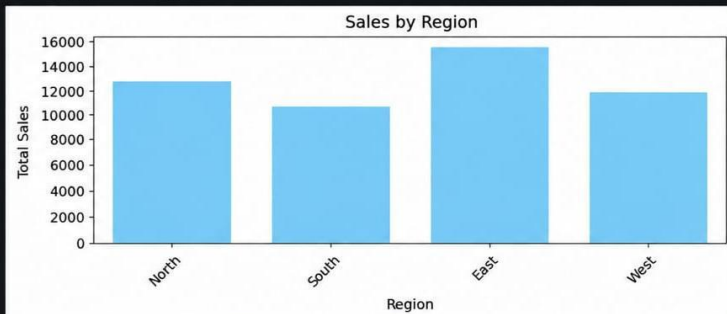
# Calculate sales by region
sales_by_region = df.groupby('Region')[['Sales']].sum().reset_index()

# Display the result
print("Sales by Region:\n", sales_by_region)

# Plot the bar chart
plt.figure(figsize=(8, 5))
plt.bar(sales_by_region['Region'], sales_by_region['Sales'], color='skyblue')
plt.title('Sales by Region')
plt.xlabel('Region')
plt.ylabel('Total Sales')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

Sales by Region:

	Region	Sales
0	North	12500
1	South	9800
2	East	15400
3	West	11200



Shows: Comparison of sales performance across different regions.

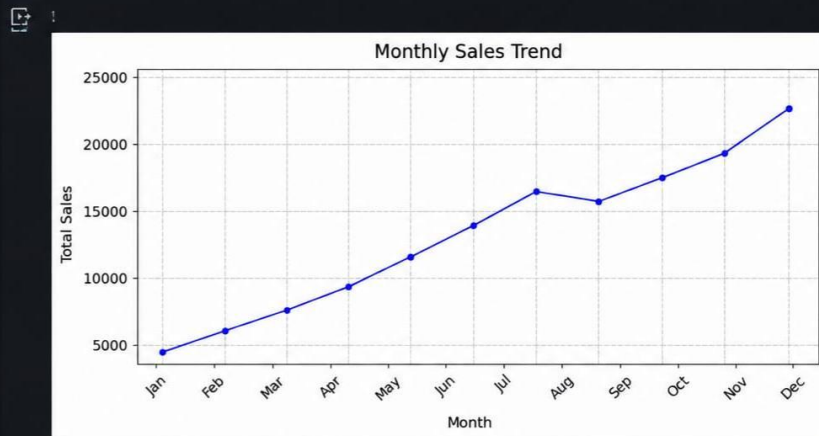
4. Monthly Sales Trend — Line Chart

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the dataset
df = pd.read_csv('/content/sales_data.csv')

# Calculate monthly sales
monthly_sales = df.groupby('Month')['Sales'].sum().reset_index()

# Plot the line chart
plt.figure(figsize=(10, 6))
plt.plot(monthly_sales['Month'], monthly_sales['Sales'], marker='o', color='blue', linewidth=2)
plt.title('Monthly Sales Trend')
plt.xlabel('Month')
plt.ylabel('Total Sales')
plt.xticks(rotation=45)
plt.grid(True, linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()
```



Shows: How sales change over time and highlights periods of higher or lower sales.

5. Discount vs Profit — Scatter Plot



Shows: The relationship between discounts offered and profit earned.